

# Quantum Computing Architecture

## Page 1: Principles of Quantum Superposition

Quantum computers exploit qubits that exist in coherent superpositions of basis states  $|0\rangle$  and  $|1\rangle$ . Quantum entanglement allows multi-qubit systems to exhibit correlated states across quantum gates.

# Quantum Algorithms: Shor and Grover

Page 2: Shors Algorithm for Prime Factorization

Shors algorithm runs in polynomial time  $O((\log N)^3)$  on a quantum computer, breaking standard RSA asymmetric cryptography. Grovers algorithm provides a quadratic speedup  $O(\sqrt{N})$  for unstructured database search.